EXTENSION BY ZERO ALONG AN OPEN IMMERSION AND ITS DEFINING PROPERTIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the isomorphisms and for the uniqueness statement.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Extension by zero j_! along an open immersion and the adjunction with j^{-1}. Same chapter, the functor j_! on abelian sheaves and its adjunction with the restriction j^{-1}. Supplies the functor and the adjunction mappings appearing in the statement.
- Stalks of sheaves and the description of extension by zero by sections with support. Same chapter, the description of the stalks of j_!F at geometric points, which vanish outside U and agree with the stalks of F on U. Supplies the identification of j_!F as the unique sheaf with the stated restriction and vanishing stalks.

Source reference. Original lemma 0F6Z: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11481–11481. Navigation pointer invoked at proof line 11481; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original proposition 03S5: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11425–11425. Navigation pointer invoked at proof line 11486; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original theorem 03PU: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3546–3546. Navigation pointer invoked at proof line 11489; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $j : U \to X$ be an open immersion of schemes. For any abelian sheaf $\mathcal{F}$ on $U_{étale}$, the adjunction mappings $j^{-1}j_*\mathcal{F} \to \mathcal{F}$ and $\mathcal{F} \to j^{-1}j_!\mathcal{F}$ are isomorphisms. In fact, $j_!\mathcal{F}$ is the unique abelian sheaf on $X_{étale}$ whose restriction to U is $\mathcal{F}$ and whose stalks at geometric points of $X \setminus U$ are zero.*

Proof. We encourage the reader to prove the first statement by working through the definitions, but here we just use that it is a special case of the very general Modules on Sites, Lemma 0F6Z. For the second statement, observe that if $\mathcal{G}$ is an abelian sheaf on $X_{étale}$ whose restriction to U is $\mathcal{F}$, then we obtain by adjointness a map $j_!\mathcal{F} \to \mathcal{G}$. This map is then an isomorphism at stalks of geometric points of U by Proposition 03S5. Thus if $\mathcal{G}$ has vanishing stalks at geometric points of $X \setminus U$, then $j_!\mathcal{F} \to \mathcal{G}$ is an isomorphism by Theorem 03PU. □